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Sardar Patel University
TYBCA (5th Sem) (NEP) Examination – 2026
US05SEBCA08|| Introduction to Artificial Intelligence



Date: 13-2-2026, Friday

Time: 10:00 am to 11:30 am

Total Marks: 25

Q.1 Multiple Choice Questions.

[05]

- 1 _____ is known as the Father of Artificial Intelligence.
a. Alan Turing b. John McCarthy c. Warren McCulloch d. Marvin Minsky
- 2 _____ structure is commonly used in semantic networks.
a. Trees b. Linked Lists c. Graphs d. Arrays
- 3 _____ search methods uses a heuristic function $h(n)$.
a. BFS b. Greedy Search c. DFS d. None of the above
- 4 _____ component of a KBS stores facts and rules.
a. User Interface b. Inference Engine
c. Knowledge Base d. Explanation Module
- 5 _____ search algorithm explores all nodes at the present depth before going deeper.-
a. Depth-First Search b. A* Search
c. Greedy Search d. Breadth-First Search

Q-2 What is a search algorithm in AI? Explain Greedy Search and A* Search algorithms.

[10]

OR

Q-2 Explain the key concepts of AI. Write a note on Expert Systems and Natural Language Processing (NLP).

[10]

Q-3 What is a Knowledge-Based System (KBS)? Describe the main components of KBS.

[10]

OR

Q-3 Explain the KBS Development Model with a Knowledge-Based Shell.

[10]
